

2(a)(i)	A – (α -) glucose ; R any qualification of glucose other than α , e.g. β -glucose B – fructose ; I hexose sugar I any qualification of fructose, e.g. α	2
2(a)(ii)	glycosidic ; R glucosidic I any qualification of glycosidic, e.g. 1–2, 1–6	1
2(a)(iii)	hydrolysis ; A acid hydrolysis	1

Question	Answer	Marks
2(b)	<i>two from:</i> sucrose is, a non-reducing sugar / non-reducing / not a reducing sugar ; A no reducing sugars present no (hydrochloric) acid used (to break down sucrose to reducing sugars) ; A sucrose not hydrolysed (to monomers / monosaccharides / reducing sugars) AVP ; (sucrose will not reduce) (blue) copper II ions to (red) copper I ions cannot donate electrons <i>idea that</i> reactive groups are not available (to react with Benedict's solution / copper ions) no (free), aldehyde / ketone, group	2
2(c)	<i>hydrostatic required once only in the answers</i> <i>four from:</i> 1 <u>diffusion</u>, into phloem <u>sieve tube</u> (element), from companion cell (through plasmodesmata) ; 2 presence of sucrose (in phloem sieve tube element), lowers water potential / makes water potential more negative ; R if in context of xylem vessels or companion cells 3 water enters (sieve tubes), by osmosis / down water potential gradient (into sieve tubes) ; 4 increase in, hydrostatic, pressure (in sieve tube elements) ; A turgor pressure 5 low hydrostatic pressure in, storage tissues / root / sink, by removal of sucrose ; 6 movement of, (phloem) sap / sucrose (in solution), down hydrostatic pressure gradient / from high to low hydrostatic pressure ; 7 <u>mass flow</u> ;	4